

LESSON 4.4

EVALUATING NUMERICAL EXPRESSIONS WITH INTEGER EXPONENTS – PART 1

LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can evaluate a numerical expression using the laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How can you use the laws of exponents to evaluate numerical expressions?

LESSON NARRATIVE

Students engage in different strategies to evaluate numerical expressions with rational bases and integer exponents using the laws of exponents.

COMPONENT SUMMARY

4.4.1 WARM-UP (~5 MIN.)

Apply laws of exponents to write equivalent expressions

4.4.3 TRY IT (5-10 MIN.)

Practice evaluating exponential expressions

4.4.5 YOUR TURN! (10 MIN.)

Practice evaluating exponential expressions

4.4.2 GUIDED INSTRUCTION (15 MIN.)

Explore different methods of evaluating exponential expressions

4.4.4 ACTIVITY (10-15 MIN.)

Card sort evaluating exponential expressions

4.4.6 WRAP-UP (~5 MIN.)

Identify and correct errors evaluating numerical expressions with rational bases and integer exponents

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Zero exponent
- Power of a power
- Power of products
- Power of quotients
- Identity exponent
- Negative exponent
- Product of powers
- Quotient of powers

MATERIALS

- Card sort

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe there is a uniform order that should always be used to evaluate expressions instead of using more efficient strategies. In 4.4.2 Guided Instruction, students reflect on two different strategies to be able to recognize that while the strategies are different, both result in the same value.

THINGS TO CONSIDER

- This lesson does not require calculators.
- Consider identifying students who struggle to complete 4.4.1 Warm-Up and pull them into a small group to provide support during the 4.4.3 Try It.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 4.4.4 Activity, students work with a partner to match each expression card with the exponent law and value card. Prompt students to first think of which cards match and then discuss their thinking with their partner. Encourage students to justify their matches by sharing the laws of exponents they applied. Prompt students to make note of when they used different methods but obtained the same answer.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 4.4.2 Guided Instruction, students analyze two students' work and describe the similarities and differences. Students discover that it is helpful to apply the laws of exponents to obtain an expression with a single base and positive exponent before evaluating the exponential expression. Students also see two different strategies when the laws of exponents are applied correctly, which result in the same value.

DIFFERENTIATE INSTRUCTION

4.4.4 Activity provides an auditory entry point as students discuss the cards they matched. Consider prompting students to use a highlighter or colored pencils to identify the parts of the expression that changed as students applied the laws of exponents to find the value.

Additional differentiation opportunities are denoted throughout the lesson guide.

Card	Expression	Value	Exponent Law(s)
a.	$\frac{1}{4^{-2}} \cdot \frac{1}{4^4}$	B	1
b.	$\frac{(-3)^0 \cdot (-3)^1}{(-3)^{-3}}$	D	2 or 4
c.	$\frac{\left(\frac{2}{3}\right)^2}{\left(\frac{2}{3}\right)^{-1}}$	C	5
d.	$\frac{(-1.5)^2}{(-1.5)^{-2} \cdot (-1.5)^4}$	E	2 or 4
e.	$3^2 \cdot \left(\frac{2}{3}\right)^{-1}$	A	3

4.4.1 WARM-UP: BELL WORK

TEACHER MOVES

Use student understanding to inform your instruction during this and further lessons. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

Consider pulling students who need additional support into a small group during the 4.4.3 Try It.



4.4.1 Warm-Up: Bell Work

- For each expression, apply one or more exponent laws to write an equivalent expression with a rational base and positive exponent.

Given Expression	Equivalent Expression
$\left(\frac{4}{5}\right)^{-5} \left(\frac{4}{5}\right)^3$	$\left(\frac{5}{4}\right)^2$
$(3^4)^{-6}$	$\left(\frac{1}{3}\right)^{24}$
$\frac{7^0}{7^{-2}}$	7^2
$\left(\frac{12}{5}\right)^{-4}$	$\left(\frac{5}{12}\right)^4$



4.4.2 Guided Instruction: Using the Laws of Exponents to Evaluate Expressions

1. Melanie and Adonis each evaluated the expression $\frac{(5)^{-2}(5)^5}{5}$. Each student's steps are shown.

Melanie	Adonis
$\frac{(5)^{-2}(5)^5}{5}$	$\frac{(5)^{-2}(5)^5}{5}$
$= \frac{5^3}{5}$	$= \frac{1}{5^2} \cdot (5)^5$
$= 5^2$	$= \frac{1}{25} \cdot 3125$
$= 25$	$= \frac{125}{5}$
	$= 25$

Discuss with your partner how Melanie's steps and Adonis' steps are similar and different from each other. Then, summarize your discussion in the table.

Similarities	Differences
<i>Both students found the value to be 25. Both started by applying a law of exponent.</i>	<i>Melanie applied product of powers first and Adonis rewrote the negative exponent first. Melanie applied exponent laws until she had the fewest number of bases, and then evaluated. Adonis evaluated the exponents and then performed the operations.</i>

Answers will vary.

4.4.2 GUIDED INSTRUCTION: USING THE LAWS OF EXPONENTS TO EVALUATE EXPRESSIONS

TEACHER MOVES

Consider prompting students to discuss whether they have a preferred strategy when comparing the two. Many students might point to Melanie's strategy as preferred because there are fewer steps, but that doesn't mean that with every expression, product of powers should be applied before negative exponents. In this case, though, it was more efficient. For other expressions, it may not be. The point is that both strategies are correct ones, and both resulted in the same value when laws were applied correctly.



When evaluating exponential expressions, it is helpful to apply the laws of exponents before evaluating.

2. Complete the steps to evaluate the expression $\frac{(6^2)^{-3}}{6^{-5}}$ by completing the description of each step and writing the resulting equivalent expression on the line that follows.

a. Apply ☐ product of power law. ☒ power of a power law. $\frac{6^{-6}}{6^{-5}}$

b. Apply ☒ quotient of power law. ☐ power of a power law. 6^{-1}

c. Evaluate. $\frac{1}{6} \text{ or } 0.1\bar{6}$

3. Consider the expression $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

a. Discuss with your partner which laws of exponents could be used to evaluate the expression.

b. Evaluate $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

25

TEACHER MOVES

Facilitate a discussion focused on the impact of applying the laws of exponents and the order in which it's done for various expressions. It will be beneficial to display an anchor chart with the laws of exponents for students to reference.

TEACHER MOVES

For question 3, teachers can choose to guide students or to have students work in partners to evaluate. If the latter, there is an opportunity to share different students' processes and discuss/compare. For example, some students may take more steps depending on the order in which they apply the laws. Focus of instruction should not be on naming the laws but rather the impact of applying them and the order in which it's done for various expressions. At this point, students will have worked with exponent laws in the 6th Grade Accelerated course and throughout this unit. There should be an anchor chart they can refer to in the room (or to the 3.1.2 Refresher in Lesson 1 in this unit if not).

4.4.3 TRY IT: EVALUATING EXPRESSIONS

ENRICHMENT

What order of steps did you take? Why did you choose this order of steps? What is another order you could have used? Try using that order and verify if the values are the same.

DIFFERENTIATE INSTRUCTION

Students may benefit from using a highlighter or colored pencil to keep track of the changes they make at each step as they apply the laws of exponents.



4.4.3 Try It: Evaluating Expressions

- Evaluate each expression. Show your work in the space provided. Then, check off the exponent laws applied to each expression from the list provided.

a. $\left(\frac{3}{4}\right)^{-2} \cdot 3$

$$\left(\frac{4}{3}\right)^2 \cdot 3$$

$$\left(\frac{4^2}{3^2}\right) \cdot 3$$

$$\left(\frac{4^2}{3}\right)$$

$$\frac{16}{3}$$

Exponent Laws Applied

- ☐ identity exponent
- ☒ negative exponent
- ☐ power of a power
- ☐ power of a product
- ☒ power of a quotient
- ☐ product of powers
- ☒ quotient of powers
- ☐ zero exponent

b. $\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$

$$\frac{(-7)^5}{(-7)^2}$$

$$(-7)^3$$

$$-343$$

Exponent Laws Applied

- ☐ identity exponent
- ☐ negative exponent
- ☐ power of a power
- ☐ power of a product
- ☐ power of a quotient
- ☐ product of powers
- ☒ quotient of powers
- ☒ zero exponent



4.4.4 Activity: Card Sort

1. Work with your partner to match each expression card with the exponent law and value cards. The exponent law cards indicate which laws of exponents can be applied to the expression before evaluating and the value cards indicate the value of the expression. Write the letter or number associated with the matching cards in the table.

Card	Expression	Value	Exponent Law(s)
a.	$\frac{1}{4^{-2}} \cdot \frac{1}{4^4}$	<i>B</i>	<i>1</i>
b.	$\frac{(-3)^0 \cdot (-3)^1}{(-3)^{-3}}$	<i>D</i>	<i>2 or 4</i>
c.	$\frac{\left(\frac{2}{3}\right)^2}{\left(\frac{2}{3}\right)^{-1}}$	<i>C</i>	<i>5</i>
d.	$\frac{(-1.5)^2}{(-1.5)^{-2} \cdot (-1.5)^4}$	<i>E</i>	<i>2 or 4</i>
e.	$3^2 \cdot \left(\frac{2}{3}\right)^{-1}$	<i>A</i>	<i>3</i>

4.4.4 ACTIVITY: CARD SORT

TEACHER MOVES

Think-Pair-Share

Prompt students to first think of which cards match and then discuss their thinking with their partner. Encourage students to justify their matches by sharing the laws of exponents they applied. If partners suggest using the same laws for a single expression, ask them to collectively determine if they can come up with another series of laws that can be applied and result in the same value.

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.

4.4.5 YOUR TURN!

ENRICHMENT

Which laws of exponents did you use? Can you use another series of steps to check your answer?

TEACHER MOVES

Consider asking students to explain why the two expressions in the middle are not equivalent to the first and last expressions.



4.4.5 Your Turn!

1. Apply the laws of exponents to evaluate each expression. Show your steps in the process.

a. $\frac{(0.6)^{-2} \cdot (0.6)^7}{(0.6)^3}$

$$\frac{(0.6)^5}{(0.6)^3}$$

$$(0.6)^2$$

$$0.36$$

b. $(-4)^3 \cdot \left(\frac{-5}{4}\right)^{-4}$

$$(-4)^3 \cdot \left(\frac{-4}{5}\right)^4$$

$$(-4)^3 \cdot \frac{(-4)^4}{5^4}$$

$$\frac{(-4)^7}{5^4}$$

$$-26.2144$$

2. Cristian was evaluating the expression $\left(\frac{5}{6}\right)^{-5} \cdot 6^2$.

- a. Circle the expressions that are equivalent to $\left(\frac{5}{6}\right)^{-5} \cdot 6^2$.

$$\frac{5^{-5}}{6^{-5}} \cdot 6^2$$

$$5^{-5} \cdot \frac{6^2}{6}$$

$$\frac{5^{-6}}{6^{-6}} \cdot 6^2$$

$$\left(\frac{6}{5}\right)^5 \cdot 6^2$$

- b. Write another equivalent expression.

Answers will vary. $\frac{6^2}{5^5}$, $\frac{6^5}{5^5} \cdot 6^2$



4.4.6 Wrap-Up: Error Analysis

- Two students each evaluated an expression. Their work is shown in the table. Each student made a mistake. Circle the step where each student made an error and describe the mistake they made.

Problem	Describe the Mistake
$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} \cdot 5^{-1}$ $= \left(\frac{1}{5}\right)^{-8} \cdot 5^{-1}$ $= 5^8 \cdot 5^{-1}$ $= 5^7$ $= 78,125$	<i>The student multiplied the exponents instead of adding them.</i>
$\left(\frac{3}{2}\right)^{-3} \cdot (2)^3$ $= \frac{2^{-3}}{3^{-3}} \cdot 2^3$ $= \frac{2^0}{3^{-3}}$ $= \frac{1}{3^{-3}}$ $= 3^3$ $= 27$	<i>The student wrote the reciprocal of $\frac{3}{2}$ but did not use the opposite exponent.</i>

4.4.6 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

QUESTIONING

Prompt students who finish early to write a letter providing advice to each student to help them avoid making the same mistake again in the future.